TERRAFORM INFRASTRUCTURE AS CODE (IaC) AUTOMATION WORKFLOW

End-to-End Azure Infrastructure Provisioning | Reusable Modules | Remote State with Blob Locking | Multi-Environment Workspaces | Shift-Left Security

TERRAFORM ARCHITECTURE
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1. GIT REPO

Azure Repos / GitHub

- Trunk-Based / Git Flow
- Isolated feature branches
- Strict branch policies
- Mandatory peer reviews
- Linked JIRA work items
- Automated PR triggers
- Zero direct commits

2. TERRAFORM CODE

Root Configuration

- Declarative HCL2 syntax
- main.tf (Orchestration)
- variables.tf (Strict types)
- outputs.tf (State exports)
- providers.tf (azurerm)
- backend.tf (Remote state)
- DRY code architecture

3. REUSABLE MODULES

Enterprise Library

- modules/vnet
- modules/subnets
- modules/nsg (Security)
- modules/linux_vm
- modules/postgresql
- modules/storage_account
- modules/key_vault

4. ENV WORKSPACES

Dev / Stage / Prod

- DEV: dev.tfvars
- STAGE: stage.tfvars
- PROD: prod.tfvars
- Workspace state isolation
- Sizing overrides per env
- Granular resource tags
- Zero hardcoded values

5. VALIDATE & PLAN

Shift-Left Gates

- terraform fmt -check
- terraform validate
- tflint (Rule verification)
- Checkov & tfsec (IaC scan)
- terraform plan -out=tfplan
- Speculative delta preview
- PR plan comment preview

6. AZURE DEPLOY

Target Cloud Infra

- terraform apply tfplan
- Service Principal (SPN)
- Azure VNet & Subnets
- NSG Security Rules
- Linux VM & Storage
- PostgreSQL Flexible
- Zero manual drift

REMOTE STATE MANAGEMENT (AZURE BLOB STORAGE BACKEND)

Backend Configuration & Concurrency Locking

- Storage Account: sttfstateprod01 | Container: tfstate (isolated keys per workspace)
- Native Azure Blob Lease provides distributed locking to prevent concurrent apply corruption

dev.terraform.tfstate
Workspace: dev
[Active] Lease: Available

stage.terraform.tfstate
Workspace: stage
[Active] Lease: Available

prod.terraform.tfstate
Workspace: prod
[Locked] Lease: Protected

Enterprise State Governance & Security Controls:

- State encrypted at rest with 256-bit AES customer-managed keys (CMK)
- Granular RBAC: Storage Blob Data Contributor access granted only to Azure DevOps Service Principal
- Engineers maintain read-only access for compliance and audit separation of duties
- Output values (VNet IDs, private IPs, DB endpoints) securely consumed by application pipelines

MULTI-ENVIRONMENT PROMOTION & GOVERNANCE FRAMEWORK

DEVELOPMENT (DEV)

Subscription: sub-telecom-dev | Workspace: dev

- Automated trigger upon Pull Request merge to main branch
- Cost-optimized sizing (Standard_B2s Linux VM, single-node PostgreSQL)
- Rapid iteration environment for functional validation and integration testing

STAGE (PRE-PRODUCTION)

Subscription: sub-telecom-stage | Workspace: stage

- High-fidelity production mirror topology and network configuration
- End-to-end performance testing, load testing and database schema dry-run
- Automated acceptance test suites verify all health checks before production sign-off

PRODUCTION (PROD)

Subscription: sub-telecom-prod | Workspace: prod

- Strict governance: Zero engineer Contributor access; apply strictly via pipeline
- Mandatory Change Advisory Board (CAB) review and multi-person approval gate
- High availability with zone-redundancy, automated backups & 24/7 telemetry